

Horizon VDI Sizer

User manual: sizing Omnisia Horizon VDI hosts on Nutanix and VMware, with persistent and non-persistent desktops and NVIDIA vGPU

Horizon VDI Sizer

NewOpen...Save...SourcesPrint / PDF

NUTANIX
Nutanix HCI
AHV with Nutanix distributed storage. A CVM runs on every node.

NUTANIX
Nutanix External Storage
AHV compute with an external array (e.g. Pure Storage FlashArray, Dell PowerFlex).

VMWARE
VMware vSAN
ESXi with vSAN. Storage runs in the hypervisor kernel.

VMWARE
VMware External Storage
ESXi with SAN or NAS storage (FC, iSCSI, NFS, NVMe-oF).

Host hardware ⓘ
The server model that runs the desktops. All hosts are assumed to be identical.

Cisco UCS catalogCustom

GenerationModelCPUSockets

M8 IntelC240 M86747P · 48c @ 2.7 GHz2

Memory

2,048 GB (32 × 64 GB)

Host memory

2048GB

☒ Hyperthreading / SMT enabled

96 physical cores · 259,200 MHz raw · 2,048 GB RAM

NVIDIA vGPU ⓘ

GPU modelGPU boards per host

L4 · 24 GB4

Ada Lovelace · Single-slot low-profile · 72 W
4 physical GPUs · 96 GB framebuffer per host
Dense, power-efficient vPC and light/mid vWS

Desktop pools ⓘ

+ Non-persistent+ Persistent

One entry per user population. Mix persistent and non-persistent pools as needed.

Contact center

PersistentNon-persistent

Instant clones: compute and vGPU are sized on peak concurrency plus spare VMs. Profiles live on a file share.

Worker profileNamed usersPeak concurrencyGolden images

Task worker1500701

vCPUVRAMCPU / uservCPU per coreIOPS / userOS disk

24GB300MHz8:11064GB

Total hosts - VMware vSAN
44
2 clusters × 22 hosts
41 for load + 3 HA/min.
Limited by GPU (vGPU slots)

"CAD engineers" uses a Q profile, which needs NVIDIA RTX vWS licenses (vPC does not cover it).

HOSTS NEEDED BY RESOURCE

CPU (MHz demand)**6**
1,315,000 MHz needed ÷ 233,280 MHz usable per host

CPU (vCPU:core ratio)**14**
1,302 cores needed ÷ 96 cores usable per host

Memory**13**
23,496 GB needed ÷ 1,817.2 GB usable per host

GPU (vGPU slots)**41**
161 GPUs needed ÷ 4 GPUs usable per host

USERS & DENSITY

Named users3,740

Peak concurrent sessions2,890

Powered-on desktops3,156

Desktops per host (all hosts up)71.7

Desktops per host (HA hosts failed)77

Usable per host96 cores · 233,280 MHz · 1,817 GB

GPUs161 needed · 176 × L4 installed

PER POOL

Pool	Concurrent	Powered on	RAM	GPUs	Disk
Contact center NP	1,050	1,155	4.6 TB	—	10.2 TB
Knowledge workers NP	1,600	1,761	13.9 TB	147 (12/GPU)	8.6 TB

Version 0.1.0 · September 2026 · For macOS and Windows

Horizon VDI Sizer · User manual · v0.1.0

1 / 19

Contents

1	What the sizer does	9	Platform overheads
2	Installing the app	10	Design assumptions
3	Quick start	11	Reading the results
4	The main window	12	How the numbers are calculated
5	Choosing the platform	13	Worked example
6	Host hardware	14	Sources & assumptions panel
7	NVIDIA vGPU	15	Saving, opening and printing
8	Desktop pools	16	Warnings and troubleshooting
		17	Limitations

A planning tool, not a guarantee

The sizer turns a workload description into a host count with a clear, repeatable method. Many of its defaults are industry rules of thumb rather than vendor-published figures, and the Sources panel labels each one (see chapter 14). Before ordering hardware, replace the profile defaults with measured data from an assessment, and validate the design with a pilot.

1 • What the sizer does

Horizon VDI Sizer calculates how many identical hosts you need to run a set of Omnissa Horizon virtual desktops. You describe three things:

- **The platform:** Nutanix HCI, Nutanix with external storage, VMware vSAN, or VMware with external storage.
- **The host:** CPU, memory and, optionally, NVIDIA GPUs. You can pick from the built-in Cisco UCS M7/M8 catalog or enter any server.
- **The users:** one or more desktop pools, each persistent or non-persistent, with a worker profile and an optional vGPU profile.

The sizer then works out how many hosts each resource needs (CPU, memory and GPU). The largest of these sets the host count. It then adds HA hosts, splits the result into clusters, and estimates storage capacity, IOPS and the Horizon management servers.

2 · Installing the app

macOS

- 1 Open `Horizon VDI Sizer-0.1.0-arm64.dmg` and drag **Horizon VDI Sizer** into **Applications**.
- 2 The app is signed ad-hoc rather than with a paid Apple Developer ID, so macOS blocks the first launch. Double-click the app; when macOS says it can't verify the app, click **Done**. Then open **System Settings** → **Privacy & Security**, scroll down to the message about the app, click **Open Anyway**, and confirm with your password or Touch ID. On macOS 14 and earlier, **right-click** → **Open** also works.
- 3 After that the app opens normally.

The DMG is built for Apple silicon (M1 and later).

Windows

- 1 Run `Horizon VDI Sizer Setup 0.1.0.exe`. The installer works on both x64 and ARM64 PCs.
- 2 The installer isn't code-signed, so Microsoft Defender SmartScreen may show "Windows protected your PC". Click **More info** → **Run anyway**.
- 3 Choose an install folder and finish the wizard. Start the app from the Start menu or the desktop shortcut.

No network or account needed

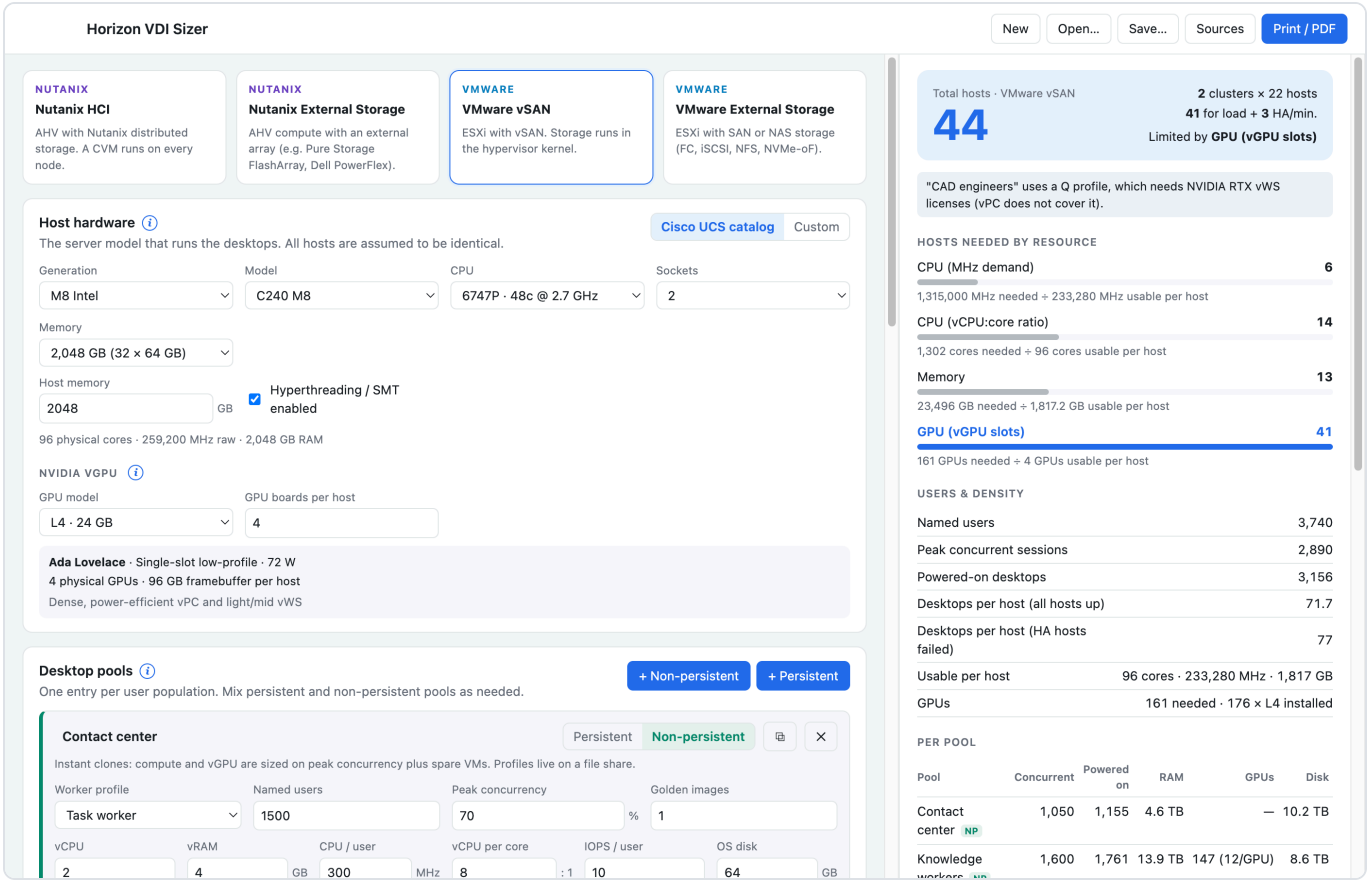
The sizer runs entirely on your computer. It doesn't connect to Intersight, vCenter, Prism or any other service. The only network activity is opening the source links in the Sources panel, which launch in your web browser.

3 · Quick start

- 1 **Pick a platform** by clicking one of the four cards at the top of the left column.
- 2 **Describe the host**. Choose *Cisco UCS catalog* and select generation, model, CPU and memory, or choose *Custom* and type the sockets, cores, clock and RAM.
- 3 **Add GPUs** (optional). Pick the GPU model and how many boards each host holds.
- 4 **Describe the users**. Edit the sample pools or add your own with + *Non-persistent* / + *Persistent*. Set named users, peak concurrency and a worker profile for each.
- 5 **Read the result** in the right column. The big number is the total host count. The highlighted bar shows which resource is limiting it.
- 6 **Save the scenario** with *Save...*, or produce a PDF of the results with *Print / PDF*.

Every change recalculates immediately. There is no "calculate" button.

4 · The main window



Inputs on the left, live results on the right.

Area	What it contains
Top bar	New (start over with the defaults), Open... / Save... (scenario files), Sources (where each default comes from), Print / PDF (prints the results panel).
Left column: inputs	Platform cards, Host hardware and NVIDIA vGPU, Desktop pools, platform overheads, and design assumptions, from top to bottom.
Right column: results	Total hosts, warnings, hosts needed per resource, users and density, per-pool breakdown, storage, and the management cluster.
 buttons	Open the Sources panel at the section you're looking at.

Hover over a field label to see its tooltip, where one exists.

5 · Choosing the platform

The platform decides how much of each host the hypervisor and storage layer keep for themselves, and how storage capacity is protected.

Platform	What changes	Storage protection options
Nutanix HCI	A Controller VM (CVM) runs on every node: 8 cores and 32 GB are reserved by default. Raw capacity per host is shown.	RF2 (2×), RF3 (3×), RF2 + erasure coding (~1.5×)
Nutanix External Storage	AHV with an external array. A smaller controller footprint (4 cores, 16 GB) is assumed. Confirm it with Nutanix.	External array (usable capacity)
VMware vSAN	vSAN memory (20 GB) and 10% CPU overhead. The SCAv2 scheduler option is available.	RAID-1 (2×), ESA RAID-5 4+1 (1.25×) or 2+1 (1.5×), OSA RAID-5 (1.33×), RAID-6 (1.5×)
VMware External Storage	ESXi only, with no storage-layer overhead. The SCAv2 scheduler option is available.	External array (usable capacity)

Switching platforms resets the overheads

When you click a different platform card, the fields in the *overheads* section reload with that platform's defaults. Edits you made there are lost, but your host, pools and assumptions stay unchanged.

6 · Host hardware

NUTANIX
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NVIDIA VGPU

 ⓘ

GPU modelGPU boards per host

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Dense, power-efficient vPC and light/mid vWS

Desktop pools

 ⓘ

Non-persistentPersistent

Cisco UCS catalog mode with an M8 C240, Xeon 6747P, 2 TB of RAM and four L4 GPUs.

Cisco UCS catalog mode

- 1 Generation:** M7, M8 Intel or M8 AMD.
- 2 Model:** the rack or blade server (e.g. C220, C240, X210c).
- 3 CPU:** the list shows cores and base clock. SKUs marked *1S only* are single-socket parts, and selecting one sets sockets to 1.
- 4 Sockets:** limited to what the model and CPU support.
- 5 Memory:** only DIMM configurations that the platform's memory guide validates are listed, for example "2,048 GB (32 × 64 GB)".

Custom mode

Type sockets, cores per socket, base clock (GHz) and host memory for any server, such as a Dell, HPE or Lenovo model or a future CPU.

Hyperthreading / SMT

When this is on, the CPU capacity per host grows by the *Hyperthreading gain* assumption (25% by default). A second thread per core doesn't double throughput.

Why base clock?

The sizer uses the CPU's base frequency, not turbo. With every core busy, turbo headroom isn't guaranteed, so the base clock gives a conservative figure.

7 • NVIDIA vGPU

NVIDIA's former "GRID" product is now called **NVIDIA vGPU software**. It has two licenses relevant to VDI. Desktop users get **vPC** (Virtual PC), which uses the **B-series** profiles. Workstation users get **RTX vWS** (Virtual Workstation), which uses the **Q-series** profiles.

Setting up GPUs

- 1 In **NVIDIA vGPU** → **GPU model**, pick the card. Its architecture, form factor, power and framebuffer appear below.
- 2 Set **GPU boards per host**. The A16 board holds four GPUs, so one A16 counts as 4 physical GPUs.
- 3 In each pool that needs graphics, set **vGPU license** to vPC (B) or RTX vWS (Q), then choose a **Profile**. The line under the profile shows how many desktops fit on one physical GPU.

GPUs and profiles in the catalog

GPU	Architecture	GPUs/board	Framebuffer / GPU	B profiles (GB)	Q profiles (GB)
RTX PRO 6000 Blackwell SE	Blackwell	1	96 GB	2, 3	3–96
L40S / L40	Ada Lovelace	1	48 GB	1, 2, 3	1–48
L4	Ada Lovelace	1	24 GB	1, 2, 3	1–24
A16	Ampere	4	16 GB	1, 2, 3	1–16
A40	Ampere	1	48 GB	1, 2, 3	1–48
A10	Ampere	1	24 GB	1, 2, 3	1–24
A2	Ampere	1	16 GB	1, 2, 3	1–16
T4 (legacy)	Turing	1	16 GB	1, 2, 3	1–16

Desktops per GPU

Density is the framebuffer divided by the profile size, capped at NVIDIA's limit of 32 vGPUs per physical GPU:

Profile	L4 (24 GB)	A16 (per board, 4 × 16 GB)	L40S (48 GB)	RTX PRO 6000 (96 GB)
1B	24	64	32	—
2B	12	32	24	32
4Q	6	16	12	24
8Q	3	8	6	12
16Q	1	4	3	6

How GPU demand is counted

Each pool is packed onto its own GPUs with a single profile size, so pools never share a physical GPU. vGPU framebuffer is allocated when a desktop powers on, so the count uses *powered-on* desktops, not concurrent sessions. vGPU desktops also need a full memory reservation, so they get no swap file.

Check the server's GPU support

The sizer doesn't know how many GPUs a particular server can hold. Before settling on *GPU boards per host*, check the server spec sheet for supported GPU counts, riser options and power supply requirements.

8 · Desktop pools

A pool is a group of users who share the same desktop type and workload. Add as many as you need. Use the  button to duplicate a pool and  to remove it. Click the pool name to rename it.

Desktop pools ⓘ

+ Non-persistent

+ Persistent

One entry per user population. Mix persistent and non-persistent pools as needed.

Contact center

PersistentNon-persistent

Instant clones: compute and vGPU are sized on peak concurrency plus spare VMs. Profiles live on a file share.

Worker profileTask worker

Named users1500

Peak concurrency70%

Golden images1

vCPU2

vRAM4 GB

CPU / user300 MHz

vCPU per core8

IOPS / user: 110

OS disk64 GB

vGPU licenseNone

Knowledge workers

PersistentNon-persistent

Instant clones: compute and vGPU are sized on peak concurrency plus spare VMs. Profiles live on a file share.

Worker profileKnowledge worker + vPC

Named users2000

Peak concurrency80%

Golden images2

vCPU2

vRAM8 GB

CPU / user450 MHz

vCPU per core6

IOPS / user: 120

OS disk80 GB

vGPU licensevPC (B)

Profile2B (2 GB)

12 per GPU

Developers

PersistentNon-persistent

Dedicated full clones: every assigned desktop keeps its disk and is usually powered on, so it is sized per named user.

Worker profilePower user

Named users200

Peak concurrency100%

vCPU4

vRAM16 GB

CPU / user1000 MHz

vCPU per core3

IOPS / user: 130

OS disk100 GB

vGPU license

Non-persistent pools have a green edge, persistent pools a purple one.

Persistent vs non-persistent

	P Persistent	NP Non-persistent
What it is	Dedicated full clones. Each user owns a desktop that keeps its disk.	Instant clones (or Nutanix clones on AHV) that reset at logoff. User data lives in a profile container.
Powered-on desktops	Every named user, because assigned desktops are normally left running	Peak concurrent users + spare VMs %, never more than named users
CPU demand	Concurrent users × MHz per user (idle desktops are assumed to use negligible CPU)	
Memory, vCPU ratio, vGPU	Powered-on desktops	
Disk	Full OS disk per named user ÷ data-reduction ratio	Delta disk per powered-on desktop, plus image replicas per cluster
Other storage	None	Profile file share per named user (FSLogix / DEM), reported separately

Horizon VDI Sizer · User manual · v0.1.0

9 / 19

Golden images	Not used	Number of images feeding the pool, which drives replica storage
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Pool fields

Field	Meaning
Worker profile	Fills in the fields below with a preset. Editing any of those fields switches the profile to <i>Custom</i> .
Named users	Everyone entitled to the pool.
Peak concurrency	The share of named users logged on at the busiest time. It defaults to 100% for persistent and 80% for non-persistent pools.
vCPU / vRAM	The desktop VM's size.
CPU / user (MHz)	Average CPU a working user actually consumes. This is the most important number to measure in an assessment.
vCPU per core	The consolidation ratio you're willing to run, e.g. 6:1 means six desktop vCPUs per physical core.
IOPS / user	Steady-state storage IO per active user.
OS disk	The desktop's system disk. For persistent pools it is stored in full; for non-persistent pools it sizes the replicas.
vGPU license / Profile	See chapter 7.

Worker profile presets

Profile	vCPU	vRAM	MHz	vCPU:core	IOPS	Disk	vGPU
Task worker	2	4 GB	300	8:1	10	64 GB	—
Knowledge worker	2	8 GB	500	6:1	20	80 GB	—
Power user	4	16 GB	1,000	3:1	30	100 GB	—
Knowledge worker + vPC	2	8 GB	450	6:1	20	80 GB	2B
Designer / engineer (vWS)	8	32 GB	2,000	2:1	40	150 GB	8Q

These presets are field rules of thumb. The knowledge-worker + vPC profile uses slightly less CPU because the GPU takes over graphics work.

9 · Platform overheads

vGPU license
RTX vWS (Q)
Profile
8Q (8 GB)
3 per GPU

VMware vSAN overheads
Reset

Defaults for the selected platform. Switching platforms resets these values.

Hypervisor RAM reserve
6 GB
vSAN memory
20 GB
CPU overhead
10 %
Per-VM memory overhead
100 MB

Datastore free-space headroom
25 %
Storage protection
ESA FTT=1 RAID-5 4+1 (1.2)
ESXi SCAv2 side-channel scheduler enabled

• vSAN memory depends on the architecture (OSA vs ESA) and the disk layout. Broadcom KB 314302 gives the OSA formula (3 GB base plus per-disk-group cost). 20 GB is a planning default.
• ESA RAID-5 adapts to cluster size: 4+1 (1.25x) on 6+ hosts, 2+1 (1.5x) on 3–5 hosts.

Design assumptions
Reset

COMPUTE

CPU utilization target
80 %
Memory utilization target
90 %
Hyperthreading gain
25 %
SCAv2 CPU penalty
25 %

Spare VMs (non-persistent)
10 %
Instant-clone parent VMs on every host (Horizon before 2306)

CLUSTERS & AVAILABILITY

HA hosts per cluster
1 N+
Max hosts per cluster
32
Remote (UAG) sessions
50 %

STORAGE

Instant-clone delta disk
5 GB
Persistent data reduction
3 :1
Profile share per user
30 GB
Logon-storm IOPS multiplier
2.5 x

Field	Effect
Hypervisor RAM reserve	Memory subtracted from every host for ESXi or AHV.
CVM / vSAN memory	Memory subtracted for the storage layer.
CVM reserved cores	(Nutanix) Physical cores removed from the desktop pool.
CPU overhead %	The share of remaining CPU lost to hypervisor and storage work.
Per-VM memory overhead	Extra memory per powered-on desktop, on top of its vRAM.
Datastore free-space headroom	Free space kept for rebuilds, snapshots and growth. Raw capacity = usable × protection ÷ (1 – headroom).
Storage protection	The capacity multiplier and the minimum number of hosts per cluster.
ESXi SCAv2 scheduler	(VMware) When enabled, CPU capacity is reduced by the <i>SCAv2 CPU penalty</i> (25% by default). Tick this if the customer's security policy requires the side-channel-aware scheduler.

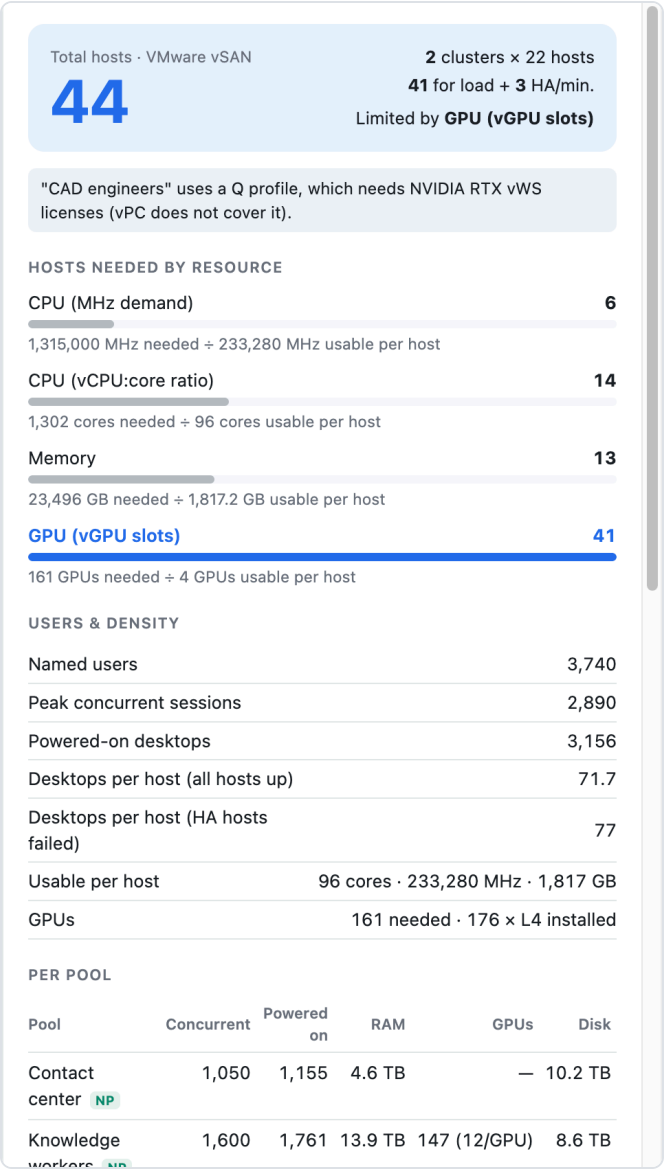
Reset restores the defaults for the selected platform.

10 • Design assumptions

The screenshot shows the Horizon VDI Sizer configuration interface. At the top, the vGPU license is set to RTX vWS (Q) with a profile of 8Q (8 GB) and 3 per GPU. Below this, the VMware vSAN overheads section is expanded, showing defaults for the selected platform. The vSAN memory is set to 20 GB, CPU overhead to 10%, and Per-VM memory overhead to 100 MB. The storage protection is set to ESA FTT=1 RAID-5 4+1 (1.2). The ESXi SCAv2 side-channel scheduler is enabled. Below this, the Design assumptions section is expanded, showing defaults for the selected platform. The CPU utilization target is 80%, Memory utilization target is 90%, Hyperthreading gain is 25%, and SCAv2 CPU penalty is 25%. The Spare VMs (non-persistent) are set to 10%. The Instant-clone parent VMs on every host (Horizon before 2306) is set to Off. The HA hosts per cluster is set to 1, Max hosts per cluster is 32, and Remote (UAG) sessions is 50%. The Instant-clone delta disk is 5 GB, Persistent data reduction is 3:1, Profile share per user is 30 GB, and Logon-storm IOPS multiplier is 2.5x.

Assumption	Default	Effect
CPU utilization target	80%	Share of host CPU you plan to use at peak.
Memory utilization target	90%	Share of host RAM you plan to allocate. VDI isn't sized with memory overcommit.
Hyperthreading gain	25%	Extra CPU capacity from SMT.
SCAv2 CPU penalty	25%	(VMware) Applied only when SCAv2 is ticked.
Spare VMs (non-persistent)	10%	Powered-on spare desktops above peak concurrency.
Instant-clone parent VMs	Off	Horizon 2306 (8.10) and later don't keep parent VMs by default. Turn this on for older versions: each image then reserves its vRAM on every host.
HA hosts per cluster	N+1	Spare hosts added to every cluster.
Max hosts per cluster	32	Upper cluster size, including the HA hosts.
Remote (UAG) sessions	50%	Share of sessions coming from outside, which sizes Unified Access Gateways.
Instant-clone delta disk	5 GB	Growth per non-persistent desktop between refreshes.
Persistent data reduction	3:1	Dedupe and compression on full-clone disks.
Profile share per user	30 GB	FSLogix / DEM container size for non-persistent users.
Logon-storm IOPS multiplier	2.5x	Peak IOPS relative to steady state.

11 · Reading the results



Total hosts, warnings and hosts by resource

The headline

Total hosts is the number to quote. To its right:

- **clusters × hosts**: how the total is split.
- **for load + HA/min.:** hosts carrying desktops, plus the extra ones. The extra count includes the HA hosts, any hosts added to reach the storage policy's minimum cluster size, and rounding from spreading load evenly across clusters.
- **Limited by**: the resource that needs the most hosts.

Hosts needed by resource

Each bar shows how many hosts one resource alone would need:

- **CPU (MHz demand)**: real CPU consumption.
- **CPU (vCPU:core ratio)**: your consolidation policy.
- **Memory**: powered-on desktops' vRAM plus overhead.
- **GPU**: physical GPUs needed ÷ GPUs per host.

The highest bar is shown in blue and sets the host count. If other bars are much lower, the design is unbalanced. For example, if GPU is the limit, adding GPUs per host reduces the host count until CPU or memory becomes the limit.

Users & density

- **Desktops per host (all hosts up):** normal operation.
- **Desktops per host (HA hosts failed):** density after losing the HA hosts. This is what the hosts must be able to carry.
- **Usable per host:** the CPU cores, MHz and GB left for desktops after all overheads.
- **GPUs:** needed vs installed across all hosts.

Per pool

Concurrent and powered-on desktops, RAM, GPUs (with density) and disk for each pool. **NP** marks non-persistent pools and **P** persistent ones.

Storage

- **Usable capacity** = replicas + persistent disks + delta disks + VM swap.
- **Raw / Array capacity** applies the protection factor and free-space headroom.
- **Raw capacity per host** (HCI only) helps choose drive configurations.
- **Profile file share** is separate storage, usually on a NAS or file service, and is not included in the datastore.
- **Steady-state** and **logon-storm peak IOPS** are for storage performance sizing.

Management cluster

Connection Servers, Unified Access Gateways, App Volumes Managers and vCenter (VMware) or Prism Central (Nutanix). The vCPU and RAM columns are totals for all VMs of that type. Put these on a separate management cluster. They are **not** included in the desktop host count.

Knowledge workers NP	1,600	1,761	13.9 TB	147 (12/GPU)	8.6 TB
Developers P	200	200	3.1 TB	—	9.6 TB
CAD engineers P	40	40	1.3 TB	14 (3/GPU)	2 TB
STORAGE					
Instant-clone replicas					896 GB
Persistent desktop disks (after data reduction)					8.5 TB
Instant-clone delta disks					14.2 TB
VM swap					7.6 TB
Usable capacity					31.2 TB
Raw capacity (ESA FTT=1 RAID-5 4+1 (1.25x), 25% free)					52 TB
Raw capacity per host					1.2 TB
Profile file share (separate)					102.5 TB
Steady-state IOPS					50,100
Logon-storm peak IOPS					125,250
MANAGEMENT CLUSTER (NOT INCLUDED IN THE HOST COUNT)					i
Component	VMs		vCPU	RAM	
Horizon Connection Server	2	8	24 GB		
1 pod x 2 (4,000 sessions each, +1 spare, max 7 per pod)					
Unified Access Gateway	2	4	8 GB		
2,000 remote sessions each, +1 spare					
App Volumes Manager	2	4	16 GB		
Only if App Volumes is used. 2 for redundancy.					
vCenter Server (desktop)	1	8	30 GB		
One per pod (Medium size)					
Planning estimate from rule-of-thumb defaults. Validate with real user telemetry (e.g. a Horizon or ControlUp assessment) and a pilot before ordering hardware.					

Per-pool, storage and management results

12 · How the numbers are calculated

Per pool

```
concurrent    = named users × peak concurrency %
powered-on    = persistent ? named users : min(named, concurrent × (1 + spare VMs %))
CPU demand    = concurrent × MHz per user
core demand   = powered-on × vCPU ÷ (vCPU per core)
RAM demand    = powered-on × (vRAM + per-VM overhead)
GPUs          = [ powered-on ÷ min([GPU framebuffer ÷ profile size], 32) ]
```

Per host

```
usable cores = sockets × cores – CVM reserved cores
usable MHz   = usable cores × base GHz × 1000
              × (1 + HT gain)           [if hyperthreading]
              × (1 – SCAv2 penalty)     [if VMware and SCAv2]
              × (1 – CPU overhead)
              × CPU target
usable RAM    = host RAM × memory target – hypervisor reserve – CVM/vSAN memory
              – Σ(images × vRAM)       [if instant-clone parents are on]
GPUs          = GPU boards × GPUs per board
```

Hosts and clusters

```
hosts for load = max( [ΣCPU ÷ usable MHz], [Σcores ÷ usable cores],
                     [ΣRAM ÷ usable RAM], [ΣGPUs ÷ GPUs per host] )
clusters       = [ hosts for load ÷ (max hosts per cluster – HA hosts) ]
hosts per cluster = max( [hosts for load ÷ clusters] + HA hosts, policy minimum )
total hosts    = clusters × hosts per cluster
```

Storage

```
replicas      = Σ(NP pools) images × OS disk × 2 × clusters (replica + internal template)
persistent    = Σ(P pools) named × OS disk ÷ data reduction
delta         = Σ(NP pools) powered-on × delta disk
swap          = Σ powered-on × vRAM (0 for vGPU desktops, which reserve all memory)
usable        = replicas + persistent + delta + swap
raw           = usable × protection factor ÷ (1 – free-space headroom)
peak IOPS     = Σ(concurrent × IOPS per user) × logon-storm multiplier
```

Horizon infrastructure

```
Pods                = [ concurrent ÷ 20,000 ]
Connection Servers  = pods × min(7, [concurrent per pod ÷ 4,000] + 1)    4 vCPU / 12 GB each
UAGs                = [ remote sessions ÷ 2,000 ] + 1                  2 vCPU / 4 GB each
App Volumes Mgr     = 2                                                2 vCPU / 8 GB each
```

13 • Worked example

The screenshots in this manual all use this scenario. Load the same numbers to follow along.

Setting	Value
Platform	VMware vSAN, ESA RAID-5 4+1, 25% free space
Host	Cisco UCS C240 M8, 2 × Xeon 6747P (48 cores, 2.7 GHz), 2,048 GB, hyperthreading on
GPU	4 × NVIDIA L4 per host

Pool	Type	Profile	Named	Concurrency	vGPU
Contact center	NP	Task worker	1,500	70%	—
Knowledge workers	NP	Knowledge + vPC, 2 images	2,000	80%	2B
Developers	P	Power user	200	100%	—
CAD engineers	P	Designer (vWS)	40	100%	8Q

Result: 44 hosts, limited by GPU

The four resources need 6 hosts (CPU MHz), 14 (vCPU ratio), 13 (memory) and **41 (GPU)**. The knowledge workers power on 1,761 desktops at 12 per L4 (147 GPUs), and the CAD engineers need 40 desktops at 3 per L4 (14 GPUs). That's $161 \text{ GPUs} \div 4 \text{ per host} = 41 \text{ hosts}$. Split across two clusters of at most 32, that's $2 \times (21 + 1 \text{ HA}) = 44 \text{ hosts}$.

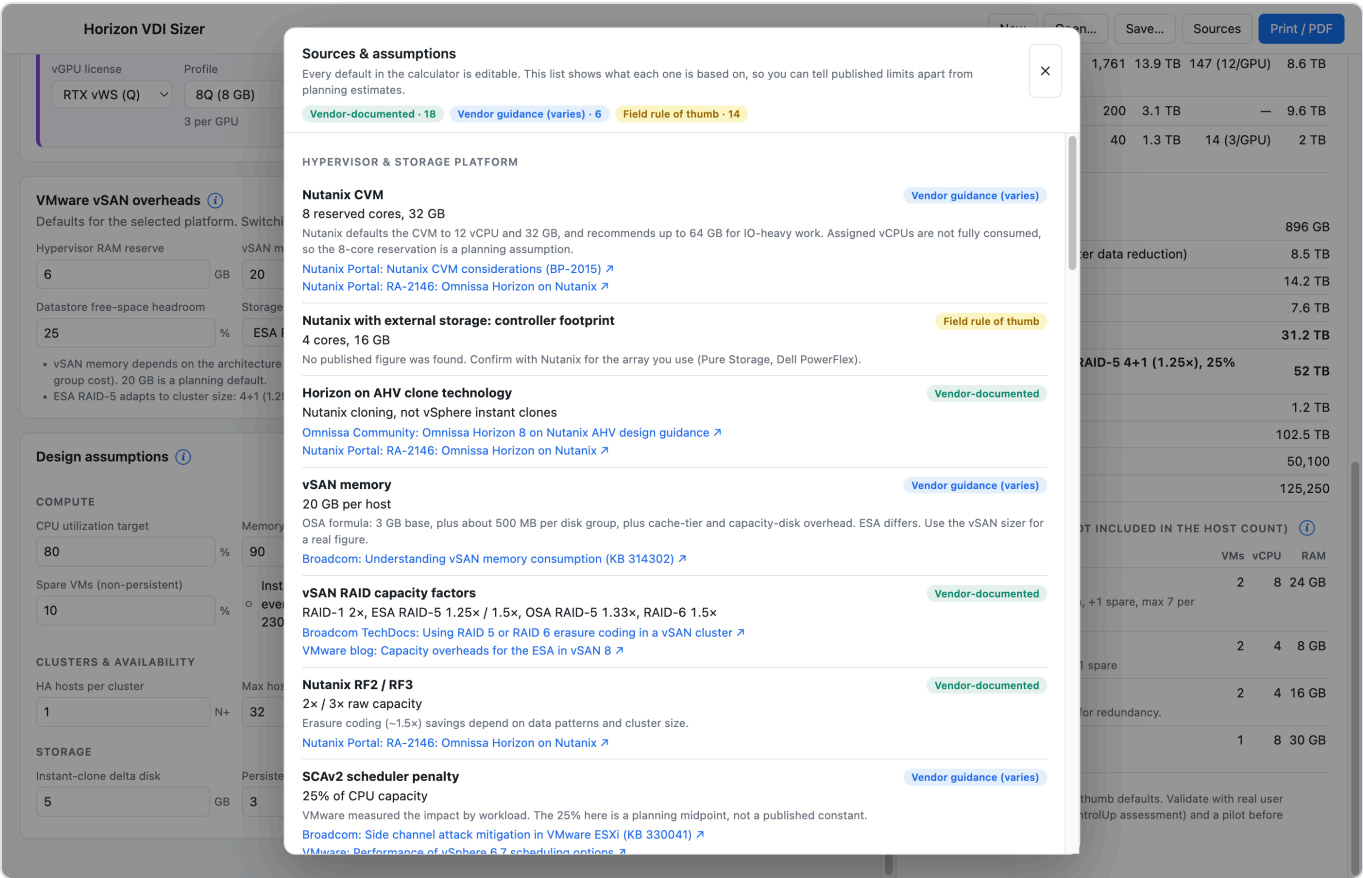
CPU and memory are mostly idle in this design, so it's a clear case for more GPUs per host. The table shows what-if results from the same engine:

Change	Hosts for load	Total hosts	Still limited by
Baseline (4 × L4 per host)	41	44	GPU
6 × L4 per host	27	28	GPU
8 × L4 per host	21	22	GPU (CPU ratio needs 14)
Knowledge workers on 1B instead of 2B (4 × L4)	22	23	GPU
2 × L40S per host instead of 4 × L4	41	44	GPU (same framebuffer per host)
Nutanix HCI, RF2, 8 × L4	21	22	GPU (CPU ratio 15, memory 14)

Takeaway

For vGPU designs, total framebuffer per host usually decides density. Change the GPU count per host and the profile size first, and only then fine-tune CPU and memory.

14 · Sources & assumptions panel



Click **Sources** in the top bar, or the **i** next to any section, to see what each default is based on. Every entry has a label:

Label	Meaning	Examples
Vendor-documented	Published by Omnissa, NVIDIA, Broadcom/VMware, Nutanix or Microsoft, with a link.	4,000 sessions per Connection Server; vGPU profile sizes; RAID factors
Vendor guidance (varies)	Vendor recommendations that depend on the configuration. Check them for your design.	Nutanix CVM sizing; vSAN memory; SCAv2 impact
Field rule of thumb	Common planning numbers with no single authoritative source. Replace them with assessment data.	MHz and IOPS per user; vCPU:core ratios; HT gain

Links open in your default web browser. Press **Esc** or click outside the panel to close it.

15 · Saving, opening and printing

- **Auto-save:** your current scenario is kept automatically and restored the next time you open the app.
- **Save...** writes the scenario to a `.json` file (e.g. `horizon-sizing-2026-09-30.json`). Keep one file per customer or design option.
- **Open...** loads a saved file. Values missing from older files are filled with the current defaults.
- **New** discards the current scenario and loads the sample defaults. You'll be asked to confirm.
- **Print / PDF** prints only the results panel. Choose *Save as PDF* in the print dialog to create a PDF you can hand to a customer.

16 · Warnings and troubleshooting

Messages appear under the headline, colored by severity: **red** means the result can't be calculated, **amber** means something needs checking, and blue is for information.

Message	What to do
Enter the host CPU...	Sockets, cores or GHz is zero. In catalog mode, pick a CPU.
No memory is left for desktops...	Host RAM is too small for the reserves (or for instant-clone parents, if enabled). Add memory or lower the reserves.
The storage controller reservation uses every core...	The CVM reserved cores exceed the host's cores.
Some pools need a vGPU, but the host has no GPU boards	Set <i>GPU boards per host</i> to 1 or more.
"..." has a vGPU profile, but no GPU is selected	Pick a GPU model, or set the pool's vGPU license to None.
...is not a valid profile on the GPU	You changed GPU model and the old profile doesn't exist on the new card. Pick a profile again.
Q profile needs NVIDIA RTX vWS licenses	Information only. Budget for vWS licenses instead of vPC.
Less than 4 GB of vRAM	Windows 11 requires at least 4 GB.
...needs at least N hosts per cluster	The storage policy requires a bigger cluster, so the sizer raised it. Choose another policy if that isn't wanted.
Non-persistent pools on AHV...	Horizon on AHV uses Nutanix cloning, so replica and delta sizing is approximate. Confirm it against the Nutanix reference architecture.
Legacy part (T4)	The T4 isn't in NVIDIA's latest vGPU release. Check availability and the software branch.

Other issues

- **macOS says the app can't be opened or can't be verified:** allow it under System Settings → Privacy & Security → Open Anyway (chapter 2). If it says the app is "damaged", delete it and download the latest DMG again.
- **The result looks too high or too low:** look at the bars under *Hosts needed by resource*. The top bar is the constraint, so check the inputs that feed it.
- **A value in the manual doesn't match the app:** the manual describes the app's defaults. Your saved scenario may have different values.

17 • Limitations

- All hosts are identical. Mixed host types or separate GPU and non-GPU clusters need separate scenarios.
- RDSH (published desktops and apps) sessions aren't modeled. Only single-user VDI desktops are.
- vGPU pools are packed per pool with one profile per GPU. Mixed-size vGPU on one GPU isn't modeled, which can overestimate GPUs slightly.
- The storage estimate assumes one datastore per cluster, and it doesn't size performance tiers, caching or network bandwidth.
- Horizon on AHV uses Nutanix cloning. The replica model is based on vSphere instant clones and is an approximation there.
- Management-cluster sizing covers the main Horizon components only. It doesn't include databases, file servers, AD or monitoring.